May 2026

Seth,

I've been going back through old files this spring and one of the things I picked up again is the trademark-novelty index we sketched out years ago and never wrote up — the `ericsilver/novelty` repo on GitHub, which until a few weeks ago contained nothing but a 22 kB README that read like an unfunded research proposal. I've now actually built the thing, and I think the surprising part is not the part I expected to be surprising. I wanted to write to you about it before deciding whether to push toward a paper.

What I went looking for

The original idea was the one we talked about: *operationalize firm innovation by reading the text trademark applicants are forced to write about themselves.* Trademarks are unusually well-incentivized text — the registrant wants the goods/services description to be broad enough to defend against future use but narrow enough to clear the examiner — so the corpus is a clean, uniform, public-domain signal of what each firm claims to be at filing time. The hypothesis I started with was "firms that file unusually-novel goods/services descriptions are firms that go on to be successful." I was expecting either a positive linear relationship between novelty and various firm outcomes, or, if the pioneer-vs-fast-follower folk wisdom held, an *inverse* linear relationship: pioneers pay the cost, fast followers reap the gain.

I set up the pipeline: USPTO Open Data Portal (16.5 M trademark filings, 5.7 M unique registrants, all 45 NICE international classes, 1985–2026), a per-class unigram + bigram dictionary, and a Kullback–Leibler surprise score for every filing computed against a 5-year look-behind reference distribution (*prospective* surprise: how unusual was the wording at the time?) and a 5-year look-ahead reference (*retrospective* surprise: how unusual is the wording in hindsight, after the industry has had five years to catch up?). The composite score $\Delta KL = KL_{\mathrm{pros}} - KL_{\mathrm{retr}}$ is positive when the future starts to look like the filing — when a novel filing gets imitated — and negative when the filing was conservative at the time and stayed conservative. I then linked the matched public-firm subset (about 14,000 USPTO owners, 1,600 with clean SEC reporting) to EDGAR financials and to monthly stock prices.

What I found that I didn't expect

The linear story is wrong. The relationship between ΔKL and survival outcomes is not monotonic; it is a clean **U-shape**, present in 38 of the 44 industries with enough data to estimate it. Both ends of the dial outperform the middle. A trademark filing in the bottom decile of ΔKL (maximally-conservative; `Beer` on a beer; `still water; sparkling water` on a water) clears registration at 48.4% and is renewed at ten years 35.7% of the time. A filing in the top decile (maximally-imitated-novel) clears at 55.5% and is renewed at 41.1%. Filings in deciles four through six — neither novel enough to claim a category nor templated enough to coast on legal-clarity grounds — do worst, 39% for both outcomes. Pooled across 8.6 M filings, the quadratic coefficient on $z(\Delta KL)^2$ is +0.009 (SE 0.0002); per-industry, 86% of industries replicate the sign, 66% at $p < 0.05$. A two-sided exponential fit $\log(\mathrm{rate}) = a + b\,|\Delta KL|$ on the bin-aggregated data gives positive slopes on both halves, $b_{\mathrm{neg}} = +0.04$ and $b_{\mathrm{pos}} = +0.05$, both $R^2 > 0.7$.

The financial pattern follows the same shape but it is dominated by the upper arm. On the matched panel of 1,597 public firms, a one-standard-deviation increase in three-year-trailing ΔKL is associated with +3.2 percentage points of gross margin (SE 0.6); on the firm-debut variant —restricting to each firm's first-ever filing so that current novelty cannot have been funded by the proceeds of past novelty — the effect attenuates to +1.7 pp but stays significant. *Stock returns are where it gets interesting.* Per σ of debut ΔKL, excess return over the S&P 500 grows from +3.6 pp at one year to +19 pp at three to +28 pp at four, then mean-reverts to +15 pp at five. The peak is several years out, not immediate; the R^2 contribution is small but real (+0.3 pp); a separate within-firm variance regression confirms the symmetric prediction that high-ΔKL firms have wider gross-margin spread, i.e. ΔKL measures something that raises both the mean and the variance of outcomes.

What the construct actually is

I started this expecting to validate "trademark novelty as a proxy for innovation." I think the construct is something slightly different and more interesting.

It is essentially uncorrelated with the dominant existing innovation proxy — patent counts ($r = +0.01$ on 174,569 matched firm-years from PatentsView) — and it is near-zero on Crunchbase total funding, weakly positive on R&D-to-revenue ($r = +0.13$), and modestly elevated on the MIT Technology Review 50 Smartest Companies list (+0.07 on-list vs. off-list). Yet on stock returns, an R^2 decomposition shows patents and ΔKL are essentially additive — their joint contribution to R^2 is the sum of the two singletons — and prospective surprise alone (mere novelty without the later-imitation marker) adds nothing whatsoever. The forward-looking "the-future-came-to-resemble-this" component is what makes ΔKL predictive.

Read together with the U-shape, the cleanest summary I have is that ΔKL measures **the smartness, and the riskiness, of the bets a firm makes when it names its products**. Average bets earn average outcomes. Bets that depart from the industry-mean vocabulary in either direction earn larger or smaller outcomes than average; the direction depends on whether the rest of the industry comes to imitate the bet, but the magnitude depends only on how far the firm departed from the centre. The five-year half-life on returns suggests the rewards take time to ferment because imitators have to recognise the new category and copy it before the originator's claim becomes financially valuable.

This is, I think, an evidentiary claim that ideas matter, literally. A firm whose trademark filings sit at the centre of the industry vocabulary distribution earns market-average margins and market-average returns. A firm whose filings depart from the centre, in either direction, has measurably different financial dynamics. That this shows up across 45 NICE classes and 1,500 matched public firms suggests it is a population-level pattern, not a quirk of software-and-pharma.

Should this be a paper?

This is the question I want to ask you.

The honest framing of the contribution, as I see it now, is:

1. A new **publicly-replicable, complete-coverage, real-time** firm-level innovation measure derived from cheap textual data with strong economic incentives behind it. The

pipeline is in a public GitHub repo with a Makefile that reproduces every figure and table from the raw USPTO and SEC sources.

2. Empirical demonstration that the relationship between novelty and firm outcomes is **not** linear, nor a clean "pioneers vs. fast followers" tradeoff, but a U-shape: *the middle of the distribution is the worst place to be.* To my knowledge no prior work in the innovation-measurement literature has documented this.

3. Demonstration that the new measure is **additive** with patent counts in predicting stock returns; the two capture different innovation channels.

4. Demonstration that the predictive horizon is several years, peaking at four, with mean reversion at five; consistent with a real category-creation/imitation cycle rather than a contemporaneous correlation artifact.

5. A reusable framework: the same Kullback–Leibler prospective/retrospective decomposition can be applied to any other dated text corpus in which the firm or individual is incentivised to be precise — 10-K business descriptions, app-store descriptions, NIH grant abstracts, Kickstarter project descriptions.

The honest concerns:

1. The construct is a *language* novelty measure, not a product or technology novelty measure. A firm that solves an old problem with an entirely new device but describes it in old vocabulary is invisible. Liquid Death's canned water is a clean example — the goods/services text is "still water; sparkling water" and ΔKL is almost zero, even though the brand was a category creator.

2. The matched-firm panel (1,600 firms) is small relative to the SEC universe ($\sim$ 13,000 firms), and the match itself is fully automated — normalised exact match plus `rapidfuzz` token-sort with cutoff 92, no manual review. There are inevitable false positives and false negatives.

3. The R^2 contributions to stock returns are real but small (under 0.5 pp incremental at any horizon). The result is statistically clean and economically meaningful per unit, but it does not explain large fractions of return variance.

4. I do not have a structural model. The narrative explanation of the U-shape is a story, not a derivation.

The questions I'd most value your read on:

- Does the U-shape result, by itself, count as a contribution? It overturns a piece of folk wisdom in the innovation literature (the "pioneers pay; fast followers win" story) by saying that both pioneers and templated followers do better than the middle. If you have seen this in print before, please point me; I have not.

- Is the right venue management/strategy (*Strategic Management Journal*, *Organization Science*), economics (*NBER Working Paper*, *AER P&P*), or information science (*Management Science*, *ICWSM* for the methodological piece)?

- Should I split the methodological contribution (KL-surprise on incentivised text) from the empirical finding (U-shape, additivity-with-patents, multi-year return cycle), or are they better together?

- What's the right second study? My current short list: a 10-K replication, a Kickstarter-vs-equity-outcome comparison, or a more careful entity-resolution pass with Compustat to enlarge the matched panel and add risk- adjusted return measures (Fama–French factors) to the stock-return regression.

The whole thing is in `github.com/ericsilver/novelty`; the longer write-up is in `paper/main.pdf`. About thirty pages. I'd rather not lose another decade between thinking about this and finishing it.

With thanks,

Eric